Fixation Chamber



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fixation chamber

What it is used for

A fixation chamber applies controlled heat, steam, or other specified conditions to fix dyes, pigments, prints, or textile finishes onto fabric. Correct process settings, fabric preparation, and exposure control help improve colour durability and finishing quality.

Main parts

- Insulated fixation chamber or enclosure
- Heating or steam-generation system
- Fabric conveyor, rack, or sample holder
- Temperature and humidity controls where fitted
- Timer, sensors, and control panel
- Ventilation system, door seals, and safety interlocks

Before you start

1. Confirm the fabric type, dye or printing process, finishing chemistry, and approved fixation recipe.
2. Read the equipment manual and chemical safety information for the materials being processed.
3. Inspect the chamber, heating or steam system, conveyor, seals, sensors, and guards for damage or leaks.
4. Verify that temperature controls, timers, ventilation, and safety interlocks function correctly.
5. Ensure that the fabric is prepared and loaded according to the process specification and that incompatible materials are not mixed.
6. Wear suitable heat-resistant gloves, eye protection, and other required PPE; ensure the work area is adequately ventilated.

How to use

1. Switch off the chamber before loading samples or adjusting internal racks and fabric guides.
2. Set the temperature, humidity or steam conditions, conveyor speed, and fixation duration according to the approved recipe.
3. Allow the chamber to reach the specified operating conditions before loading, where required by the process.
4. Arrange the fabric evenly on the designated rack or conveyor without folds, overlapping that restricts treatment, or contact with hot surfaces.
5. Start the fixation cycle and verify that the fabric moves or remains positioned as intended.

6. Monitor temperature, humidity or steam indicators, exposure time, and fabric condition during operation.
7. Check for uneven heating, condensation problems, fabric scorching, colour bleeding, or unusual odours.
8. Stop the process if the temperature exceeds limits, steam escapes unexpectedly, ventilation fails, or the fabric shows signs of damage.
9. Do not open the chamber during operation unless the equipment and approved procedure specifically allow safe access.
10. At the end of the cycle, stop heating or steam supply and follow the prescribed cooling and ventilation procedure.
11. Open the chamber only when conditions are safe, then remove the fabric using appropriate heat-resistant protection.
12. Allow the fabric to cool or condition as required and inspect colour fixation, appearance, and other specified quality parameters.
13. Record the recipe, operating readings, batch identification, and inspection results.
14. Switch off and isolate power before cleaning or maintenance, and clean the chamber according to the manufacturer's instructions.

⚠ Safety

- Only trained personnel should operate the fixation chamber and select process settings.
- Treat chamber surfaces, steam, and processed fabric as hot until confirmed safe to handle.
- Do not exceed the equipment's rated temperature or pressure limits.
- Keep guards, door interlocks, and pressure-relief devices in place and functional.
- Maintain ventilation when processing dyes, coatings, or chemicals that may release hazardous vapours.
- Use chemical-compatible PPE when handling treated fabric or process chemicals.
- Isolate electrical and steam supplies before maintenance, cleaning near hazardous components, or repairing leaks.

Care & storage

- Remove textile residue and deposits after the chamber has cooled, using approved cleaning methods.
- Inspect seals, sensors, heating elements, steam lines, and conveyor components regularly.
- Check temperature control accuracy and arrange calibration at the required intervals.
- Keep ventilation paths clear and inspect the exhaust system for obstructions.
- Arrange servicing if temperature uniformity deteriorates, interlocks fail, or fabric quality becomes inconsistent.

Common mistakes to avoid

- Using incorrect temperature, humidity, steam conditions, or fixation time for the textile process.
- Overloading the chamber and preventing uniform treatment.
- Opening the chamber before it has cooled or reached a safe condition.
- Ignoring uneven colour fixation, scorching, chemical odours, or faulty ventilation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.